

Problems

①

- ① A discrete source emits one of 5 symbols one every millisecond. The symbol probabilities are $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16} + \frac{1}{16}$ resp. Find the source entropy and information rate.

Sol: Entropy (H) = $\sum_{i=1}^5 P_i \log_2 \frac{1}{P_i}$

$$= \frac{1}{2} \log_2 \frac{1}{(\frac{1}{2})} + \frac{1}{4} \log_2 \frac{1}{(\frac{1}{4})} + \frac{1}{8} \log_2 \frac{1}{(\frac{1}{8})} + \frac{1}{16} \log_2 \frac{1}{(\frac{1}{16})} + \frac{1}{16} \log_2 \frac{1}{(\frac{1}{16})}$$
$$= 0.5 + 0.5 + 0.375 + 0.25 + 0.25$$

$$H = 1.875 \text{ bits/symbol}$$

$$\text{Information rate} = r \cdot H$$

$$= 1000 \cdot 1.875$$

$$R = 1875 \text{ bits/sec.}$$

- ② Find the entropy of a source that emits one of 3 symbols A, B, C in a statistically independent sequence with probabilities $\frac{1}{2}, \frac{1}{4} + \frac{1}{4}$ resp.

Sol: $H = \sum P_i \log_2 \frac{1}{P_i}$

$$\Rightarrow \frac{1}{2} \log_2 \frac{1}{(\frac{1}{2})} + \frac{1}{4} \log_2 \frac{1}{(\frac{1}{4})} + \frac{1}{4} \log_2 \frac{1}{(\frac{1}{4})}$$
$$= 0.5 + 0.5 + 0.5 = 1.5 \text{ bits/symbol.}$$

- ③ A discrete source emits one of 6 symbols once every millisecond. The symbol probabilities are $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$ resp. Find the source entropy and Information rate.

Sol

$$H = \sum P_i \log \frac{1}{P_i}$$

$$= 0.5 + 0.5 + 0.375 + 0.25 + 0.15625 + 0.15625$$

$$= 1.9375 \text{ bit/symbol}$$

$$\text{Information rate (R)} = r H$$

$$= 1000 \times 1.9375$$

$$= 1937.5 \text{ bit/sec}$$

- ④ Consider a discrete memoryless source with source alphabet $S = \{s_0, s_1, s_2\}$ and source statistics $\{0.7, 0.15, 0.15\}$. Calculate the entropy of source.

Sol

$$H = \sum P_i \log \frac{1}{P_i} \Rightarrow - \sum P_i \log P_i$$

$$= 0.7 \log_2 \frac{1}{0.7} + 0.15 \log_2 \frac{1}{0.15} + 0.15 \log_2 \frac{1}{0.15}$$

$$= 0.3602 + 0.4105 + 0.4105$$

$$H = 1.1812 \text{ bit/symbol}$$

⑥. The o/p of an information source consists of 150 symbols, 32 of which are of probability $\frac{1}{64}$ + remaining with a probability $\frac{1}{236}$. ③

The source emits 2000 symbols/sec. Assuming that the ~~source emits~~ symbols are chosen independently. Find the average information rate of this source.

Sol Find; $R = r \cdot H$.

$$H = \left[\frac{1}{64} \log_2 \frac{1}{(\frac{1}{64})} \right] \times 32 + \left[\frac{1}{236} \log_2 \frac{1}{(\frac{1}{236})} \right] \times 118$$

$$= 0.09375 \times 32 + 0.03340 \times 118$$

$$= 3 + 3.9413$$

$$H = 6.9413 \text{ bits/symbol.}$$

$$\text{Given } r = 2000 \text{ symbols/sec.}$$

$$R = r \cdot H$$

$$= 2000 \times 6.9413$$

$$\boxed{R = 13,882.6 \text{ bits/sec}}$$

4. The information morse code uses a sequence of dots and dashes to transmit letters of english alphabets. The dash is represented by a current pulse of duration 3 units. and the dot has a duration of 1 unit.

The probability of occurrence of a dash is $\frac{1}{3}$ of the probability of occurrence of a dot.

(a) Calculate the information Content of a dot + dash

(b) Calculate the average information in the dot, dash code

(c) Assume that the dot lasts 1msec, which is the same time interval as the ~~same~~ same b/n symbols. Find the average rate of information transmission

Sol: $P_{\text{dot}} + P_{\text{dash}} = 1$

$$P_{\text{dash}} = \frac{1}{3} P_{\text{dot}} \Rightarrow \boxed{P_{\text{dot}} = 3P_{\text{dash}}}$$

$$\frac{1}{3} P_{\text{dot}} + P_{\text{dot}} = 1$$

$$P_{\text{dot}} = \frac{3}{4}$$

$$P_{\text{dash}} = \frac{1}{4}$$

$$(a) \cdot T_{\text{dot}} = \log_2 \frac{1}{\left(\frac{2}{4}\right)} = \log_2 \frac{4}{3} = 0.4150 \quad \frac{\log_2 (4/3)}{\log_2 2}$$

$$T_{\text{dash}} = \log_2 4 = 2 \text{ bits}$$

$$(b) \quad H = \sum p_i \log \frac{1}{p_i}$$

$$= \frac{3}{4} \log_2 \frac{4}{3} + \frac{1}{4} \log_2 4$$

$$= 0.311278 + 0.5$$

$$H = 0.811278 \text{ bits/symbol.}$$

$$(c) \quad T_{\text{dot}} = 1 \text{ msec}, \quad T_{\text{dash}} = 3 \text{ msec}, \quad T_{\text{symbol}} = 1 \text{ msec.}$$

$$T = 3 \times 1 \text{ msec} + 1 \times 3 \text{ msec} + 4 \times 1 \text{ msec.}$$

$$= 10 \text{ msec}$$

$$r = 4 \text{ symbol/10 msec.} = 400 \text{ symbols/sec.}$$

$$I(R) = r \cdot H$$

$$= 400 \times 0.811278$$

$$= 324.5112 \text{ bits/sec.}$$

⑧ Consider a telegraph source having 2 symbols ⑥
 dot and dash. The dot duration is 0.25 sec and
 the dash duration is 3 times of the dot duration.
 The probability of the dots occurring is twice
 that of dash and time between symbols is
 0.2 sec. Calculate the information rate of the
 telegraph, source producing the string of
 1200 symbols.

Sol

$$\text{dash} = 'p'$$

$$\text{dot} = '2p'$$

$$\Rightarrow p + 2p = 1$$

$$\boxed{p = \frac{1}{3}}$$

$$P_{\text{dash}} = \frac{1}{3} ; P_{\text{dot}} = 2\left[\frac{1}{3}\right] = \frac{2}{3}$$

$$\text{Entropy: } (H) = \frac{1}{3} \log_2 3 + \frac{2}{3} \log_2 \frac{3}{2}$$

$$= 0.5283 + 0.4206$$

$$= 0.9489 \text{ bits/symbol.}$$

$$\text{Calculate Rate: } (R) = T_{\text{dot}} = 0.25 \text{ sec.}$$

$$T_{\text{dash}} = 3(\uparrow \text{dot}) = 0.75$$

$$\text{Time b/n Symbols} = 0.2 \text{ sec.}$$

$$T = T_{\text{dot}} + T_{\text{dash}} + [1200 * T_{\text{symbol}}]$$

$$\text{Number of dots} = \frac{2}{3} \times 1200 = 800$$

$$\text{Number of dash} = \frac{1}{3} \times 1200 = 400$$

$$T = [800 \times 0.2] + [400 \times 0.7] + [1200 \times 0.2]$$

$$= 160 + 280 + 240$$

$$= 680 \text{ sec.}$$

$$r = \frac{1200}{680} = 1.764$$

$$\boxed{r = 1.764 \text{ symb/sec.}}$$

$$R = r \cdot H$$

$$= 1.764 \times 0.9183$$

$$\boxed{R = 1.619 \text{ bits/sec}}$$

① Example:

Let $S = \{s_1, s_2, s_3\}$

o/p = $\{s_1, s_1, s_1, s_2, s_1, s_3, s_1, s_1, s_1, s_1, s_2, s_3\}$

$N = 11$ } Total no of symbols
available in the o/p }

For this source.

$$P = \{7/11, 2/11, 2/11\}$$

i.e., 7 times of s_1

2 times of s_2

2 times of s_3

$$\text{Entropy} = \sum_{i=1}^M p_i \log_2 \frac{1}{p_i}$$

$$= \frac{7}{11} \log_2 \left(\frac{1}{7/11} \right) + \frac{2}{11} \log_2 \left(\frac{1}{2/11} \right) + \frac{2}{11} \log_2 \left(\frac{1}{2/11} \right)$$

$$= \frac{7}{11} \log_2 \left(\frac{11}{7} \right) + \frac{2}{11} \log_2 \left(\frac{11}{2} \right) + \frac{2}{11} \log_2 \left(\frac{11}{2} \right)$$

$$\Rightarrow \frac{7}{11} \log_2 \left(\frac{11}{7} \right) + 2 \left(\frac{2}{11} \log_2 \left(\frac{11}{2} \right) \right)$$

$$H = 1.3092 \text{ bits/symbol.}$$

2) A Source produces one of 4 possible symbols, having probabilities $P(x_1) = \frac{1}{2}$, $P(x_2) = \frac{1}{4}$, $P(x_3) = P(x_4) = \frac{1}{8}$. Obtain the information content of each of these symbols.

Solⁿ:
$$I(x_i) = \log_2 \left(\frac{1}{P(x_i)} \right)$$

$$I(x_1) = \log_2 \left(\frac{1}{(1/2)} \right) = \log_2(2) = 1 \text{ bit}$$

$$I(x_2) = \log_2 \left(\frac{1}{(1/4)} \right) = \log_2(2^2) = 2 \text{ bits}$$

$$I(x_3) = \log_2 \left(\frac{1}{(1/8)} \right) = \log_2(2^3) = 3 \text{ bits}$$

$$I(x_4) = \log_2 \left(\frac{1}{(1/8)} \right) = \log_2(2^3) = 3 \text{ bits}$$

Max^m length of bits = 3.

Min^m length of bits = 1.

← X →

a binary channel, of '0' occur with prob $\frac{1}{4}$ & occurs with probability of $\frac{3}{4}$. Then calculate amount of information caused by each bit.

here, given. bit '0' has $P(x_1) = \frac{1}{4}$ and

bit '1' has $P(x_2) = \frac{3}{4}$

Amount of information is given as,

$$I(x_i) = \log_2 \frac{1}{P(x_i)}$$

$$P(x_1) = \frac{1}{4} ;$$

$$I(x_1) = \log_2 4 = \frac{\log_{10} 4}{\log_{10} 2} = 2 \text{ bits}$$

$$P(x_2) = \frac{3}{4} \quad \log\left(\frac{1}{3/4}\right)$$

$$I(x_2) = \log_2 \left(\frac{4}{3}\right) = \frac{\log_{10} (4/3)}{\log_{10} 2} = 0.415 \text{ bits}$$

Extensions of Sources

Consider a memoryless source 'S' with alphabet $\{0, 1\}$ and emission probabilities. $p(0) = 0.2$ & $p(1) = 0.8$.

⇒ If the source is extended then, the alphabet will be

$$S = \{00, 01, 10, 11\}$$

$$p(00) = p(0)p(0) = (0.2)(0.2) = 0.04$$

$$p(01) = p(0)p(1) = (0.2)(0.8) = 0.16$$

$$p(10) = p(1)p(0) = (0.8)(0.2) = 0.16$$

$$p(11) = p(1)p(1) = (0.8)(0.8) = 0.64$$

⇒ If extended 3rd time, the alphabet will be

$$S = \{000, 001, 010, 011, 100, 101, 110, 111\}$$

$$p(000) = 0.008$$

$$p(001) = 0.032$$

$$p(010) = 0.032$$

$$p(011) = 0.128$$

$$p(100) = 0.032$$

$$p(101) = 0.128$$

$$p(110) = 0.128$$

$$p(111) = 0.512$$

—x—

$$H(A/b_i) = \sum p(a_i/b_i) \log \frac{1}{p(a_i/b_i)} = \frac{12}{15} \log_2 \frac{1}{(12/15)} + \frac{1}{15} \log_2 \frac{1}{(1/15)}$$

$$=$$

$$H(A/b) = p(a_i/b) \log \frac{1}{p(a_i/b)}$$

0.6 marks. ~~on Matrix Information~~

(11)

Consider the binary information channel fully specified.

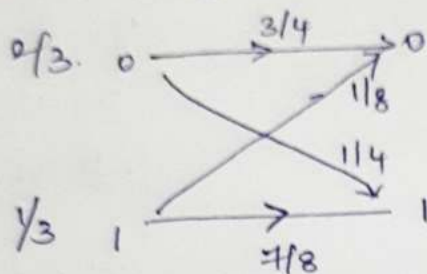
$$P = \begin{bmatrix} 3/4 & 1/4 \\ 1/8 & 7/8 \end{bmatrix}$$

$$P(a=0) = 2/3$$

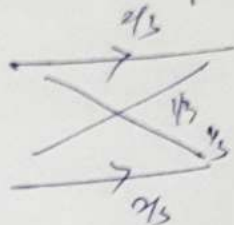
$$P(a=1) = 1/3$$

represent in diagram

Graphical diagram:-



(a) o/p probabilities



$$P(b=0) = P(b=0|a=0)P(a=0) + P(b=0|a=1)P(a=1)$$

$$= \frac{2}{3} \left(\frac{3}{4} \right) + \frac{1}{3} \left(\frac{1}{8} \right)$$

$$= \left(\frac{3}{4} \times \frac{2}{3} \right) + \left(\frac{1}{8} \times \frac{1}{3} \right) = \frac{13}{24}$$

$$\Rightarrow \frac{13}{24}$$

$$P(b=1) = 1 - P(b=0)$$

$$= 1 - \frac{13}{24} = \frac{11}{24}$$

$$P(a=0|b=0) = \frac{P(b=0|a=0)P(a=0)}{P(b=0)} = \frac{\left(\frac{3}{4} \right) \left(\frac{2}{3} \right)}{\frac{13}{24}} = \frac{12}{13}$$

$$P(a=1|b=0) = 1 - P(a=0|b=0) = \frac{1}{13}$$

$$P(a=0|b=1) = \frac{P(b=1|a=0)P(a=0)}{P(b=1)} = \frac{\left(\frac{1}{4} \right) \left(\frac{2}{3} \right)}{\frac{11}{24}} = \frac{4}{11}$$

$$P(a=1|b=1) = 1 - P(a=0|b=1) = \frac{7}{11}$$

PTO.

Consider a binary information channel.

$$H(A|B) = \sum p(a|b_j) \log_2 \frac{1}{p(a|b_j)}$$

$$H(A) = \frac{2}{3} \log_2 \frac{3}{2} + \frac{1}{3} \log_2 3$$

$$H(A) = 0.918$$

$$\Rightarrow H(A|b=0) = 0.391 \quad H(X|A=0) = 0.241$$

$$H(A|b_j) = \sum p(a|b_j) \log_2 \frac{1}{p(a|b_j)} \quad \begin{aligned} p(a=0|b=0) &= \frac{12}{13} \\ p(a=1|b=0) &= \frac{1}{13} \end{aligned} \quad H(X|b=1) = 0.831$$

$$H(A|b=1) = 0.946$$

$$\text{i.e., } p(a=1|b=1) = \frac{7}{11}$$

$$p(a=0|b=1) = \frac{4}{11}$$

$$H(A|b=1) \geq H(A)$$

$$H(A|b=0) < H(A)$$

$$\begin{aligned} \text{If we calculate } H(A|B) &= \sum_{j=1}^2 P(b_j) H(A|b_j) \\ &= P(b=0) H(A|b=0) + P(b=1) H(A|b=1) \\ &= \left(\frac{11}{24}\right) 0.391 + \left(\frac{13}{24}\right) 0.946 \\ &= 0.645 \approx 0.691 \end{aligned}$$

$$\text{Thus } H(A|B) < H(A)$$

$H(A|B)$ measures the amount of uncertainty. In other words, we have about the i/p A, when observing o/p B.

For the channel, we know, $H(A) = 0.918$

$$H(A|B) = 0.645$$

(IV)

Let's calculate $H(B)$

$$H(B) = \sum p(b) \log_2(b)$$

$$= \frac{11}{24} \log_2 \frac{24}{11} + \frac{13}{24} \log_2 \frac{24}{13}$$

$$H(B) = 0.995$$

From this, we can calculate $I(A; B)$
($H(A)$, $H(B)$, $H(A|B)$) $H(B|A)$
 $H(A, B)$

we know that $I(A; B) = H(A) - H(A|B)$
 $= 0.918 - 0.645$

$$I(A; B) = 0.273$$

$$H(B|A) = H(B) - I(A; B)$$
$$= 0.995 - 0.273$$

$$H(B|A) = 0.722$$

$$H(A, B) = H(A) + H(B) - I(A; B)$$
$$= 0.918 + 0.995 - 0.273$$

$$H(A, B) = 1.640$$

which can also be projected as.

vi

$$p(a_i | b_j, c_k) = p(a_i | b_j) \quad \forall i, j, k$$

Mutual Information with cascade of channel.

AB with BC is $I(A; C)$

$$I(A; B) = H(A) - H(A|B)$$

$$I(A; C) = H(A) - H(A|C)$$

If channel is noiseless $I(A; B) = I(A; C)$

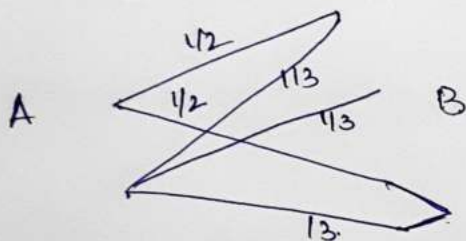
$$p(a|b) = p(a|b, c)$$

$$p(a, b, c) = p(b, c) p(a|b, c)$$

Ex: Consider a channel AB cascaded with BC, find

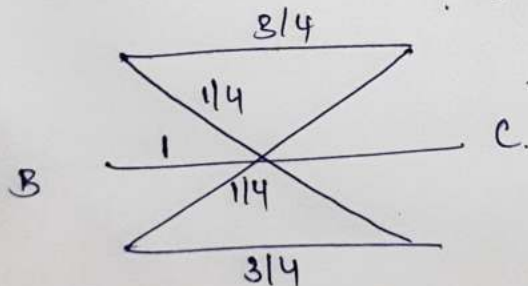
AC. Calculate PAc matrix, ~~also prove~~ Justify the channel.

AB



$$P(AB) = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$$

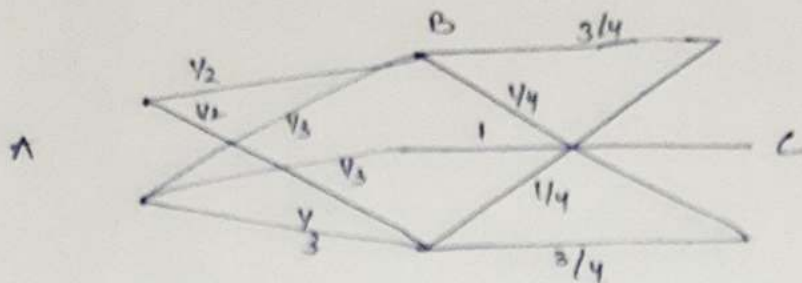
BC



$$P(BC) = \begin{bmatrix} 3/4 & 0 & 1/4 \\ 0 & 1 & 0 \\ 1/4 & 0 & 3/4 \end{bmatrix}$$

Castaded channel. AC

Q



$$P(AC) = P_{AB} P_{BC}$$

$$= \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{3}{4} & 0 & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} \end{bmatrix}$$

$$= \begin{bmatrix} \left(\frac{1}{2} \cdot \frac{3}{4}\right) + 0 + \left(\frac{1}{2} \cdot \frac{1}{4}\right) & \frac{1}{2} \cdot 0 + 0 + 0 \cdot \frac{1}{2} & \left(\frac{1}{2} \cdot \frac{1}{4}\right) + 0 + \left(\frac{1}{2} \cdot \frac{3}{4}\right) \\ \left(\frac{1}{3} \cdot \frac{3}{4}\right) + 0 + \left(\frac{1}{3} \cdot \frac{1}{4}\right) & \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0 & \left(\frac{1}{3} \cdot \frac{1}{4}\right) + 0 + \frac{1}{3} \cdot \frac{3}{4} \end{bmatrix}$$

$$= \begin{bmatrix} \left(\frac{3}{8} + \frac{1}{8}\right) & 0 & \left(\frac{1}{8} + \frac{3}{8}\right) \\ \left(\frac{3}{12} + \frac{1}{12}\right) & \frac{1}{3} & \frac{1}{12} + \frac{3}{12} \end{bmatrix} = \begin{bmatrix} \frac{4}{8} & 0 & \frac{4}{8} \\ \frac{4}{12} & \frac{1}{3} & \frac{4}{12} \end{bmatrix}$$

$$P_{AC} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

$\therefore P_{AB} = P_{AC}$; Hence. the channel is noiseless.